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# Zain Ul Abidin

## Education

2022 - 2026 **B.E. Computer Systems Engineering**, *Sukkur IBA University*, Sukkur, Pakistan
*CGPA: 3.80/4.00, Rank: 2/39*
Thesis: Machine Learning-Based Subsurface Defect Detection in Carbon Fiber Reinforced Polymer Using Optical Pulsed Thermography (Supervisor: Dr. Junaid Ahmed) **[Thesis] [Code]**

## Research Interests

Embedded Machine Learning (TinyML), Edge AI Hardware Deployment, Robotics Perception, Sensor Fusion, Digital System Design, Computational Signal Processing

## Technical Expertise

| | |
|---|---|
| Hardware & Embedded | Raspberry Pi, Arduino (Uno), ESP32, Microcontrollers, Physical Interfacing (Vision/Thermal Sensors, Actuators), Digital Logic Design (FSMs) |
| Signals & Math | Digital Signal Processing (DSP), Discrete/Wavelet Transforms, Spatial-Temporal Filtering, Statistical Feature Extraction, Applied Algorithms |
| Machine Learning | Deep Learning, Computer Vision, Image Segmentation, Object Detection, PyTorch, TensorFlow (Lite), OpenCV, Predictive Modeling |
| Programming | Python, C/C++, MATLAB, Verilog, SQL |

## Publications

### Published

2026 H. Memon, **Z. U. Abidin**, J. Ahmed, A. Siddiqua. "Debond Detection in CFRP Structures using Optical Pulse Thermography and Deep Learning." *5th International Conference on Computing, Mathematics and Engineering Technologies (iCoMET)*, IEEE. **[DOI]**

2025 **Z. U. Abidin**, Haseeb, H. Memon, M. Younis, J. Ahmed. "Mitigation of DDoS Attacks with Machine Learning, Deep Learning, and Transformers." *International Journal of Multidisciplinary Conference Proceedings (IJMCP)*. **[DOI]**

### Journal Preprints & Under Review

2026 **Z. U. Abidin**, H. Memon, J. Ahmed. "Computational Depth Measurement in Thermographic Video: Overcoming Spatial Overfitting via Spatio-Temporal Decoupling." **[arXiv]**

2026 **Z. U. Abidin**, H. Memon, J. Ahmed. "Three-Channel Thermographic Fusion for Architecture-Independent Defect Detection and Segmentation in Carbon Fibre Reinforced Polymer Composites Using Deep Learning." **[SSRN]**

2025 **Z. U. Abidin**, J. Ahmed, D. Wang, G. Tian. "Metal Defect Detection in Pulsed Thermographic Imaging via Temporal Eigen Projection and Robust Low-Rank Tensor Completion Model." **[ResearchGate]**

2025 J. Ahmed, **Z. U. Abidin**, G. Tian, D. Wang. "Eddy Current Pulsed Thermography based Efficient Defect Detection in Metals via Keyframe Extraction and Accelerated Sparse Decomposition." **[ResearchGate]**

### Conference Presentations

2026 **Paper Presentation:** "Debond Detection in CFRP Structures using Optical Pulse Thermography and Deep Learning" | *5th Int. Conf. on Computing, Mathematics and Engineering Technologies (iCoMET 2026)*

2025 **Paper Presentation:** "Mitigation of DDoS Attacks with Machine Learning, Deep Learning, and Transformers" | *Int. Conf. on Innovations in Science, Computing & Technology (ICISCT'25)*

## Applied ML & Hardware Engineering Projects

Academic **X-ray Prohibited Item Detection in Airport Inspection**
Engineered deep learning segmentation architectures to precisely identify prohibited items within highly cluttered X-ray security scans. **[Code]**

Academic **Hand Gesture Recognition and Voice Conversion**
Developed a lightweight CNN-SVM classification model on hand-landmark coordinates, converting detected sign language configurations into synthesized edge-hardware speech. **[Code]**

Academic **AI-Based Android Controlled Vision Robot**
Built an ESP32-CAM robotics platform interfacing motor actuators with TensorFlow Lite for efficient, on-device edge object detection without cloud computing dependencies. **[Code]**

Academic **Smart Traffic Light Controller (Mealy FSM)**
Designed a sensor-driven adaptive traffic controller using a Mealy Finite State Machine in Verilog, processing physical vehicle states to dynamically regulate signals for FPGA deployment. **[Code]**

Academic **Real-Time Speech Enhancement**
Implemented a DSP pipeline for noisy acoustic signals, estimating background noise to apply Spectral Subtraction and frequency-domain Wiener Filtering for improved speech intelligibility. **[Code]**

Academic **Real-Time Speech Emotion Recognition**
Built an RNN-based emotion recognition system, applying data augmentation techniques (noise injection, time-shifting) to improve feature extraction robustness in noisy acoustic environments. **[Code]**

## Experience

Aug 2025 - May 2026 **Independent Researcher (Perception & NDE)**, *Sukkur IBA University*
Designed spatio-temporal tensor decomposition algorithms for thermographic signals and engineered deep learning pipelines to segment physical anomalies, leading to multiple academic publications.

Jul 2025 - Aug 2025 **Artificial Intelligence Intern**, *Mirai School of Technology*, Remote
Architected automated data parsing pipelines in Python and deployed robust API integrations for real-time data ingestion, ensuring high-fidelity structures for downstream models.

Jun 2025 - Jul 2025 **Data Engineering Intern**, *Sukkur Electric Power Company (SEPCO)*, Sukkur
Engineered a Python/SQLite spatial monitoring system analyzing utility metrics, implementing GIS-based filtering algorithms for automated sensor anomaly detection that reduced manual verification by 40%.

## Honors & Awards

2024 **Prime Minister's Laptop Award** - National merit-based distinction for top academic rank.

2022 - 2026 **Sindh Educational Endowment Fund (SEEF)** - Full merit-based undergraduate scholarship.

## Certifications

Mathematics for ML and Data Science | *DeepLearning.AI*

Machine Learning Specialization | *Stanford / Coursera*

Deep Learning Specialization: Advanced AI, Hands-on Lab | *Udemy*

Google IT Support | *Google*

## References

**Dr. Junaid Ahmed** - Assistant Professor, Sukkur IBA University (j.bhatti@iba-suk.edu.pk)

**Dr. Kashif Hussain** - Assistant Professor, Sukkur IBA University (kashif.memon@iba-suk.edu.pk)

**Dr. Ghulam Abbas** - Assistant Professor, Sukkur IBA University (ghulam.abbas@iba-suk.edu.pk)